ZAINAB ABBAS HUSAINI

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PROFILE SUMMARY

Aspiring AI/ML Engineer with hands-on experience building and deploying machine learning and NLP systems, including transformer-based sentiment classification, semantic similarity search, and Flask-based ML web applications. Proficient in Python, PyTorch, and Scikit-learn, with a strong foundation in data structures, algorithms, and object-oriented programming. Passionate about solving real-world problems through data-driven approaches and eager to contribute to impactful AI/ML projects.

EDUCATION

B.Tech CSE, specializing in Artificial Intelligence & Machine Learning July 2023 – May 2027
Shri Ramswaroop Memorial College of Engineering and Management, Lucknow | CGPA: 7.7

Class XII (CBSE) 2022 – 2023
Lucknow Public School, Lucknow | Percentage: 80%

SKILLS

Technical Skills: Python, C++, HTML, CSS, Machine Learning, Deep Learning, NLP, Data Structures & Algorithms, Object-Oriented Programming (OOP), PyTorch, Hugging Face Transformers, Scikit-learn, Sentence-Transformers, NumPy, Pandas, Matplotlib, Seaborn, Flask, Streamlit, Git, Jupyter Notebook, VS Code

Soft Skills: Analytical, Self-Motivated, Collaborative, Problem Solver, Detail-Oriented, Continuous Learner

PROJECTS

AI-Powered Answer Verification & Explainability System

- Built a Streamlit application that verifies free-text answers against live Wikipedia content using semantic similarity rather than keyword matching.
- Implemented Sentence-Transformer (all-MiniLM-L6-v2) embeddings and cosine similarity to classify answers as Correct, Partially Correct, or Incorrect, with a rule-based heuristic to flag likely AI-generated text.
- Added sentence-level explainability by matching answer sentences to source sentences, giving users transparent, auditable verdicts.

Tech Stack: *Python, Streamlit, Sentence-Transformers, NLTK, Wikipedia REST API*

DistilBERT IMDb Sentiment Classifier

- Fine-tuned a pretrained DistilBERT (distilbert-base-uncased) transformer model for binary sentiment classification on 50,000 IMDb movie reviews, achieving 86.4% test accuracy.
- Built an end-to-end NLP pipeline covering tokenization, model fine-tuning, and evaluation using Hugging Face Transformers and PyTorch.
- Used Hugging Face's Trainer API to streamline the training and evaluation workflow on GPU-accelerated infrastructure.

Tech Stack: *Python, PyTorch, Hugging Face Transformers*

Algerian Forest Fires — Fire Weather Index (FWI) Prediction

- Built a regression pipeline to predict Fire Weather Index (FWI) from weather and fuel-moisture data across two Algerian regions (244 records).
- Performed data cleaning, exploratory data analysis, and correlation analysis; compared Linear, Ridge, and Lasso regression models, using LassoCV for automated regularization tuning.
- Deployed the trained Ridge regression model as a Flask web application with an HTML form for real-time FWI prediction.

Tech Stack: *Python, Pandas, NumPy, Scikit-learn, Flask, Matplotlib, Seaborn*

Moodify — Mood-Based Song Recommendation System

- Built a sentiment-analysis-based system that recommends songs based on user mood, using NLP techniques to classify emotions from text input.
- Mapped detected sentiments to curated song recommendations, automating personalized music suggestions and improving user experience.

CERTIFICATIONS & ACHIEVEMENTS

- Learn C++ Programming – Beginner to Advance (Deep Dive in C++), Udemy (2023).
- Gained strong understanding of C++ and OOP concepts — including inheritance, polymorphism, encapsulation, and abstraction — through practical coding exercises.